UNNI CO2 SMARTIFICATION — REV A

Two-sheet cleanup: standard power symbols, grid-aligned wiring, explicit local connectivity.

USB + Power

File: 01_usb_power.kicad_sch

ESP32-C6 + Auxiliary

File: 02_esp_aux.kicad_sch

Simulation Harness

File: 03_simulation_harness.kicad_sch

Unni CO2 Smartification

Sheet: /

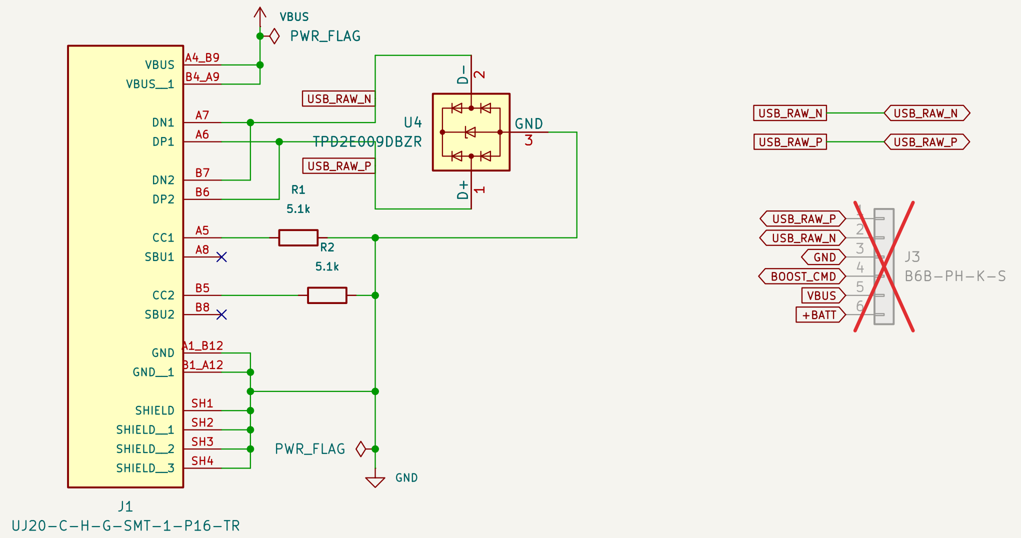
File: unni-smartification-c6.kicad_sch

Title: Unni CO2 Smartification — ESP32-C6 Rev A

Size: A3	Date: 2026-08-19	**Rev: A-v29-validated**
KiCad E.D.A. 10.99.0-3277-gdfba7e335e		Id: 1/4

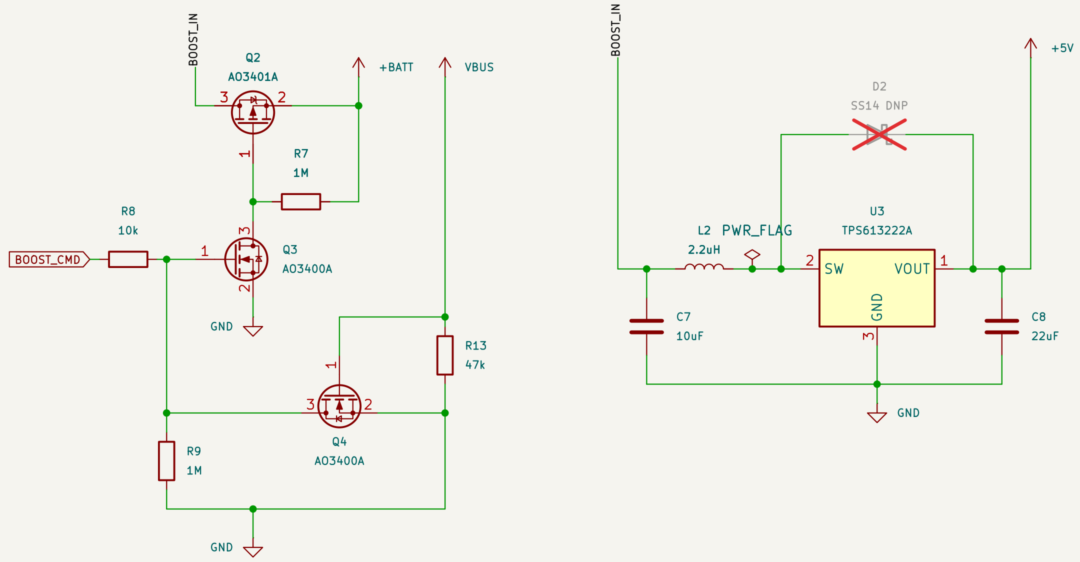
USB + POWER

All placement/wiring is on 50 mil (1.27 mm) grid. Standard KiCad symbols used where available.

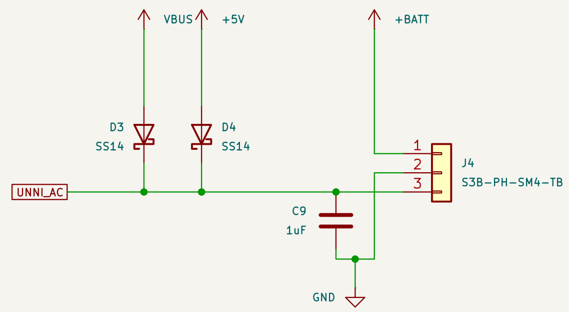


MCP73831 is on the ESP/battery board: R3 = 5.1 kΩ sets ~196 mA; STAT is monitored by the ESP. Battery NTC remains firmware-supervised.

FORCED-AWAKE 5V



UNNI POWER OUTPUTS



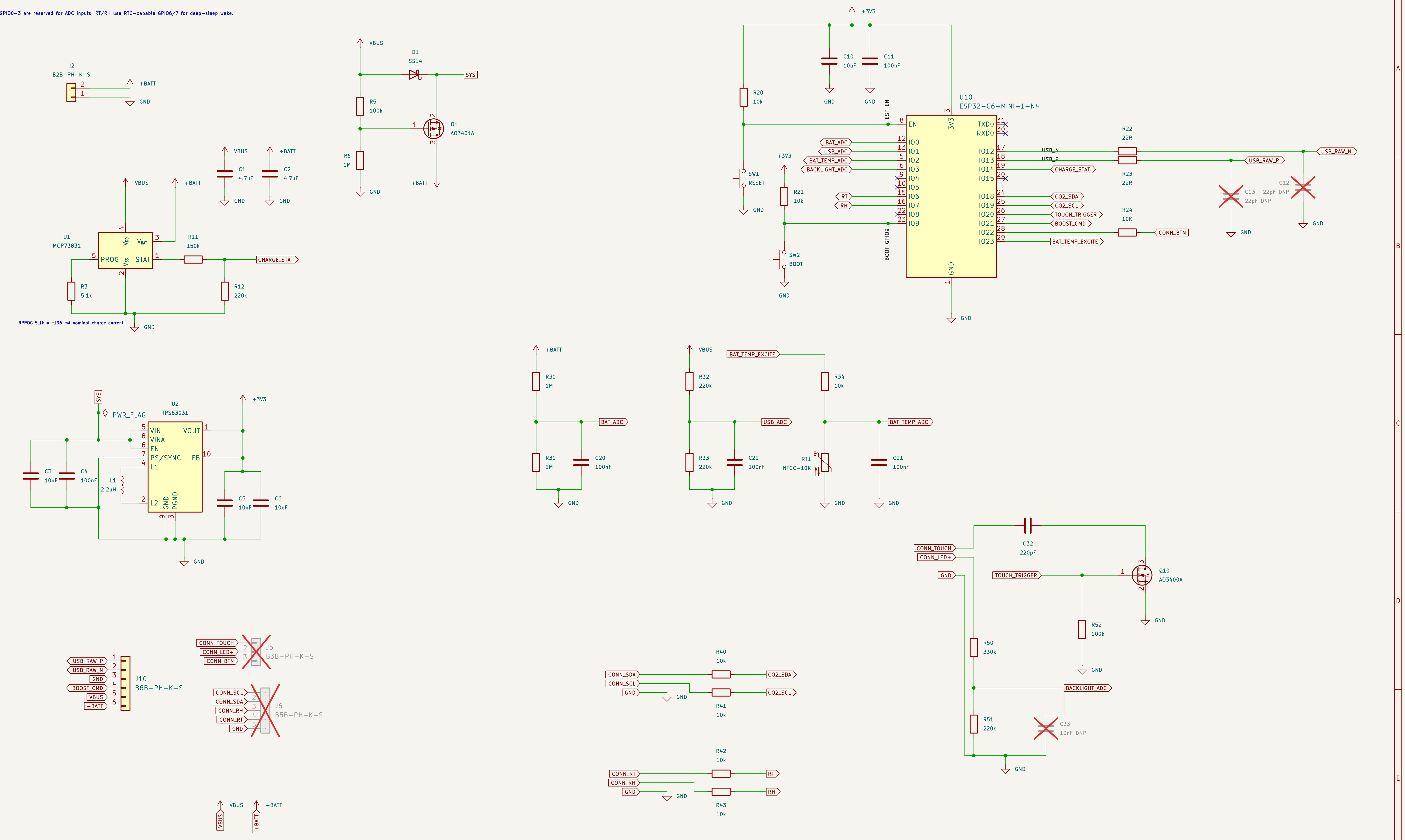
Unni CO2 Smartification
Sheet: /USB + Power/
File: 01_usb_power.kicad_sch

Title: USB + Power

Size: A3 | Date: 2026-08-19 | Rev: A-v29-validated
KiCad E.D.A. 10.99.0-3277-gdfba7e335e | Id: 2/4

ESP32-C6 + AUXILIARY CIRCUITS

GPIO0–3 are reserved for ADC inputs; RT/RH use RTC-capable GPIO6/7 for deep-sleep wake.



Unni CO2 Smartification

Sheet: /ESP32-C6 + Auxiliary/

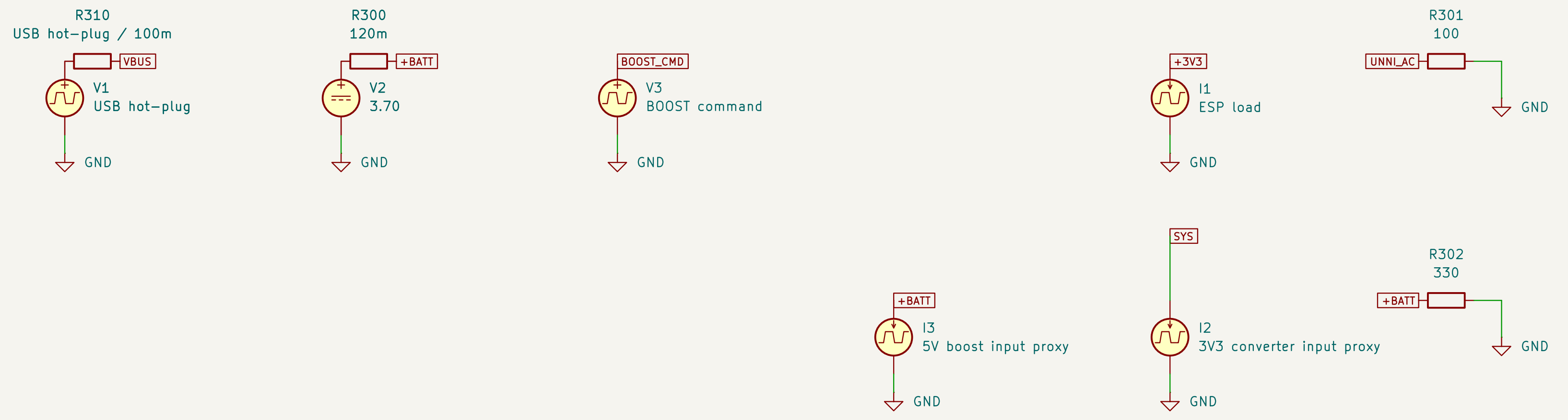
File: 02_esp_aux.kicad_sch

Title: ESP32-C6 + Auxiliary Circuits

Size: A3	Date: 2026-08-19	Rev: A-v29-validated
KiCad E.D.A. 10.99.0-3277-gdfba7e335e		Id: 3/4

SIMULATION HARNESS — excluded from BOM/PCB

Scenario: battery–only → forced awake → USB hot–plug while BOOST_CMD remains asserted → autonomous charging → USB removal. Battery model includes 120 mΩ ESR. Battery model includes 120 mΩ ESR. I2 is the 3V3–converter input–power proxy. TPS613222A input power/discharge is modeled internally; legacy I3 is excluded from simulation.



Unni CO2 Smartification	
Sheet: /Simulation Harness/	
File: 03_simulation_harness.kicad_sch	
Title: Simulation Harness	
Size: A3 Date: 2026–08–19	**Rev: A–v29–validated**
KiCad E.D.A. 10.99.0–3277–gdfba7e335e	Id: 4/4